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## **Assessing Diagenetic Variability and Its Role in Reservoir Heterogeneity: Implications for Enhanced Oil Recovery in the Tuba Formation, Kuwait**

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### **Abstract**

This study evaluates the impact of diagenetic variability on reservoir heterogeneity in the Tuba Formation (Upper Cretaceous), Kuwait. It aims to characterize how cementation, dolomitization, dissolution, and compaction influence porosity, permeability, and fluid flow properties. The study also investigates how these variations affect Enhanced Oil Recovery (EOR) strategies.

The research integrates core analysis, thin-section petrography, scanning electron microscopy (SEM), Energy Dispersive X-ray Spectroscopy (EDS), Mercury Injection Capillary Pressure (MICP) testing, and well log analysis to examine the diagenetic features and their effect on reservoir quality. A conceptual 3D model was developed to visualize the distribution of facies and diagenetic zones.

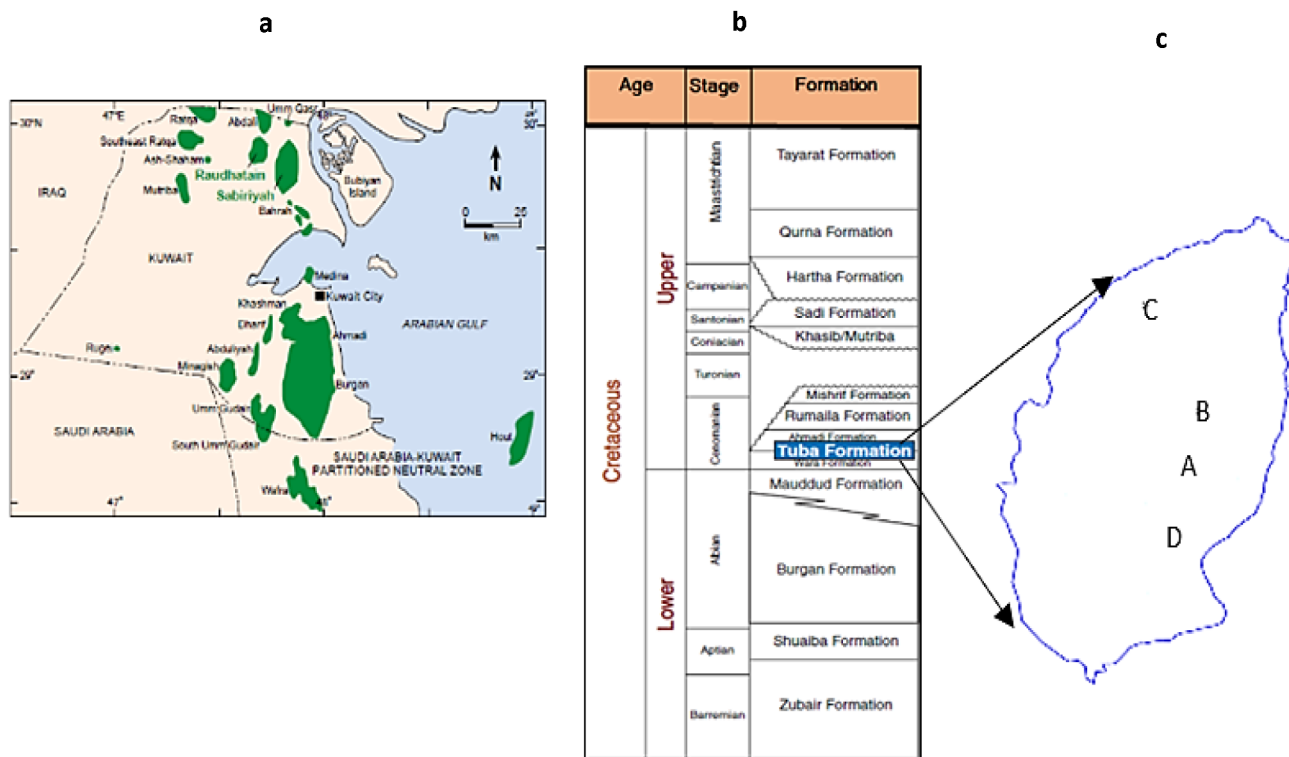
The study revealed that diagenetic variability significantly impacts the heterogeneity of the Tuba Formation. High-permeability zones (e.g., dolomitized grainstones and dissolution-enhanced facies) exhibit intercrystalline and vuggy porosity, while low-permeability zones (e.g., cemented mudstones and compacted wackestone) show pore occlusion and low permeability. MICP results confirm that dolomitized and dissolution-enhanced zones have higher permeability (up to 500 mD), while cemented zones exhibit low permeability (<0.1 mD). 3D modeling visualized how facies and diagenetic features control fluid flow in the formation. The study's findings suggest that EOR strategies must account for diagenetic heterogeneity, with targeted interventions (e.g., fracturing or acidizing) in cemented zones, while optimizing water flooding or CO<sub>2</sub> injection in more permeable streaks.

This study used a different approach to integrating diagenetic analysis with 3D modeling to assess reservoir heterogeneity, offering insights into optimizing EOR techniques in tight carbonate reservoirs.

### **Introduction**

Carbonate reservoirs are well known for their complex porosity systems and heterogeneity arising from depositional facies variability and overprinting diagenetic processes. The Upper Cretaceous Tuba Formation

in Kuwait exemplifies such heterogeneity – it is a tight, unconventional carbonate reservoir characterized by lateral and vertical variations in facies and significant diagenetic overprint (Sunil, 2024). Regionally, the Tuba Formation of north Kuwait is bounded by impermeable shales, being deposited above the Wara Shale and sealed by the overlying Ahmadi Shale (Ranjan, 2018). Internally, it is divided into twelve facies association units based on reflecting depositional cycles (Farhan, 2024). The formation's depositional environment is interpreted as a low-energy carbonate ramp, with facies ranging from high-energy shoal grainstones to low-energy mudstone and shale intervals (Sunil, 2024). However, subsequent diagenesis has further modified the pore structure and connectivity within these layers, leading to a highly heterogeneous developed reservoir with stratified flow units (Farhan, 2024).



**Figure 1—Stratigraphic position and geographic distribution of the Tuba Formation in Kuwait.(a)Regional map showing the distribution of oil fields and structural features across Kuwait and surrounding parts of the Arabian Gulf. (Modified after Zaidi (2009)).(b) Generalized Cretaceous stratigraphic column of Kuwait showing the Tuba Formation within the Upper Cretaceous succession, positioned between the Wara and Ahmadi Formations.(c) Outline map of field and approximate position of oil wells**

Previous work has highlighted that the Tuba reservoir contains multiple facies associations and corresponding flow units, including high-quality layers (skeletal grainstones/packstones with well-connected pores) and tight intervals (argillaceous wackestone/mudstone with cemented or isolated pores). These flow units often act as reservoirs, baffles, or barrier layers that can impede vertical and lateral fluid flow (Farhan, 2024).

In the Tuba Formation, detailed core-based facies and diagenetic analyses have revealed multiple generations of diagenetic overprint that critically influence current reservoir quality (Sunil, 2024). Furthermore, any Enhanced Oil Recovery (EOR) strategy must contend with the extreme permeability contrasts and compartmentalization caused by diagenesis. Conventional floods in heterogeneous carbonates often suffer from early breakthrough and poor sweep efficiency due to channeling through high-perm streaks (Zheng,2010). Therefore, a tailored approach that accounts for diagenetic heterogeneity is required to improve oil recovery in the Tuba reservoir.

This study aims to assess the diagenetic variability within the Tuba Formation and evaluate its role in creating reservoir heterogeneity, which has direct implications for EOR. We integrate geological and petrophysical data from core to reservoir scale to characterize the types and distribution of diagenetic alterations e.g., type of cements, dolomitization, dissolution vugs, stylolite's, fractures, quantify their impact on porosity and permeability using laboratory measurements and delineate diagenetic flow units or zones within the reservoir that can act as preferential flow conduits or barriers. A conceptual three-dimensional model is constructed to visualize facies and diagenetic zonation across the field, showing how these factors contribute to reservoir compartmentalization. The implications of the observed heterogeneity for EOR are discussed, with recommended strategies

## Scope and Methodology

The study followed an integrated workflow to characterize the Tuba Formation's reservoir heterogeneity and diagenetic features. Representative samples from identified facies from four Tuba wells (Well-A, B, C, D) were subjected to thin-section petrography, revealing pore types and diagenetic overprints through transmitted light analysis. To gain micro-scale insights, Scanning Electron Microscopy (SEM) with Energy Dispersive X-ray Spectroscopy (EDS) was employed on selected plugs, allowing for the identification of pore-throat geometry, micro porosity, and elemental composition of cements. Mercury Injection Capillary Pressure (MICP) analysis was carried out on 6–7 samples, providing quantitative data on pore throat size distribution, capillary entry pressure, and saturation profiles, which were instrumental in linking diagenetic facies to flow behavior. Routine core analysis provided porosity and permeability measurements later validated against wireline log data. Cross-plots of porosity versus permeability, facies, and diagenetic type enabled visualization of petrophysical trends and the influence of diagenetic overprints. A conceptual 3D static model was developed to depict the spatial distribution of facies and diagenetic zones.

## Results

### Facies Framework

The Tuba Formation was deposited on a low-gradient ramp system, exhibiting significant lateral and vertical facies variations. This depositional framework records multiple transgressive-regressive cycles, evidenced by key flooding surfaces. (Farhan, et.al, 2024).

The Tuba Formation broadly contains ten principal facies grouped into twelve facies associations, reflecting a ramp-type depositional setting. Coarse-grained facies, including peloidal skeletal grainstone, skeletal packstone, and bioturbated intraclastic wackestone, typically exhibit macropore-dominated pore systems with primary interconnected and secondary moldic or dissolution-enhanced porosity. (Sunil, 2024). These facies are dominant in the lower and upper depths. Fine-grained facies, comprising argillaceous skeletal mudstone and wackestone, display matrix-supported textures and limited porosity, confined mainly to micro pores. These finer facies are distributed throughout, particularly in the middle and upper depths.

This depositional layering established a basic reservoir architecture of alternating higher and lower quality layers even before diagenesis. Several regionally traceable flooding surfaces bound these facies packages, which correlate across wells and define the flow units observed (as reservoir, baffle, or barrier intervals in previous stratigraphic studies).

### Diagenetic Alterations: Types and Distribution

However, diagenetic processes have overprinted this primary fabric, often amplifying it. The heterogeneity allows for significant and minor compartmentalization.



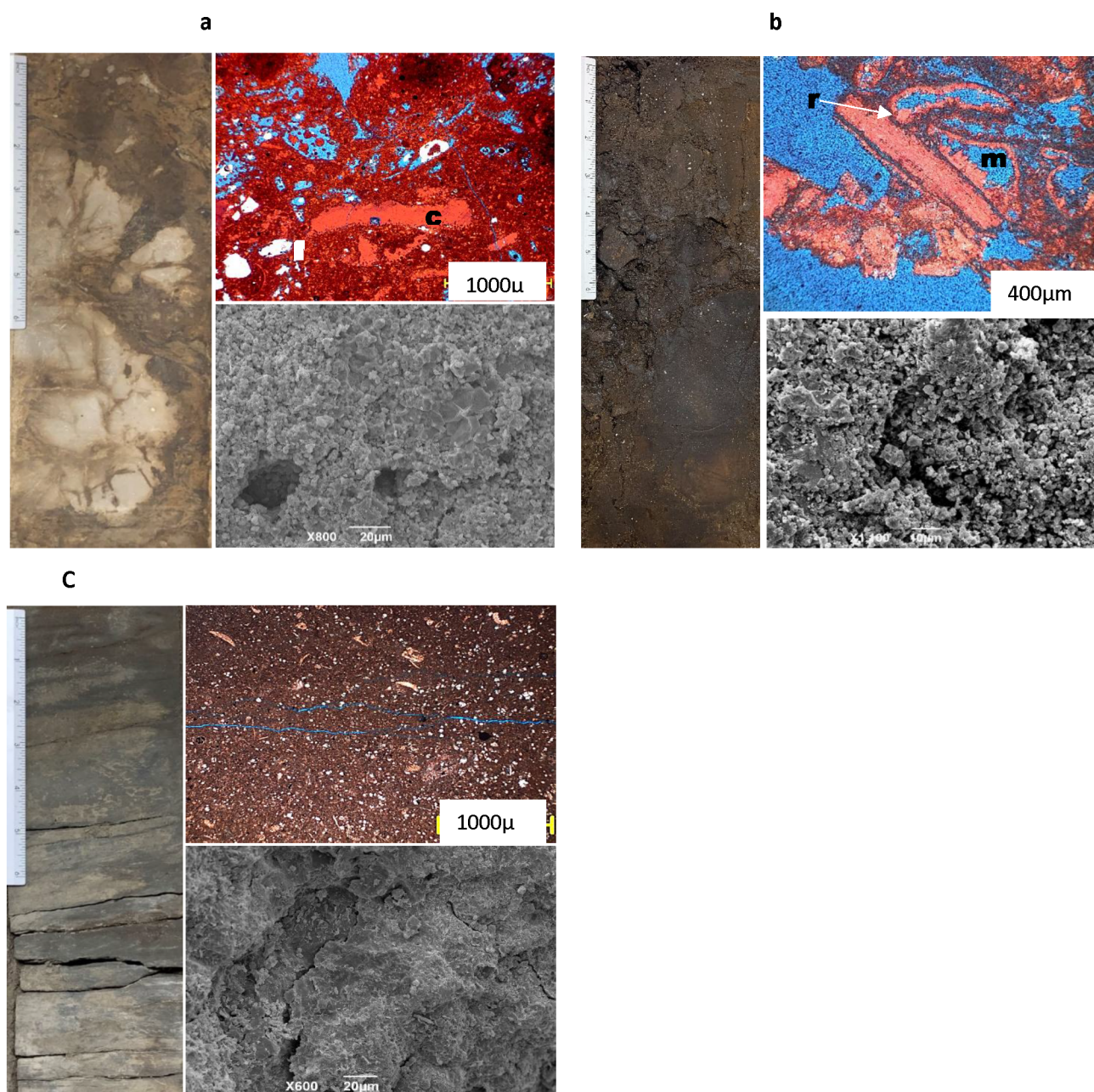


Figure 2—Representative core(left), thin section(top), and SEM images(bottom) from the Tuba Formation illustrating diagenetic variability across high (HDZ), moderate (MDZ), and low (LDZ) diagenetic zones. a- HDZ features (left)- Argillaceous skeletal packstone large nodular cementation with interconnected fractures (top): photomicrograph showing matrix rich texture with abundant moldic(m) partially filled with late-stage calcite (c) cement; the dissolution-enhanced pore space (blue dye indicates porosity) (bottom): SEM image showing moldic pores(m) and developed macro calcite (c) crystals b – MDZ features (top)- Skeletal grainstone with less cementation (top): photomicrograph showing preserved moldic pores (m), isopatchous grain rimming cements (r) developed partially. (bottom): dominant primary and moldic pores c – LDZ features (top)- Shale with planar fractures (top)/ (bottom) mudstone with tight mud matrix

## Cementation

In the Tuba Formation, pervasive calcite cementation especially matrix-replacive coarse spar is observed across all facies, most intense in argillaceous skeletal wackestone and mudstone. These intervals exhibit near-complete pore occlusion, resulting in extremely low permeability (<0.01–0.1 mD). Coarser facies like skeletal grainstones and packstones show equant and rim cements (r), indicating early marine cementation. Skeletal wackestone and packstone often contain nodular (n) cements and clay seams, with moderate



secondary pore connectivity and medium permeability (<15 mD). Vuggy porosity is commonly occluded by later cements, forming persistent diagenetic barriers across wells

### **Compaction and Pressure-Solution**

All core samples show signs of mechanical and chemical compaction to varying degrees. Grains are tightly packed and occasionally fractured in grain-supported fabrics, a sign of mechanical compaction under overburden stress. Additionally, the nodular cements associated with skeletal wackestone and packstone are fractured internally.

They are highly distributed in finer facies, such as skeletal wackestone and argillaceous mudstone. They appear to be most associated with nodular cementation, acting as a binding seam in these cementations. Overall, compaction reduced porosity and, in tandem with cementation, has created laterally extensive diagenetic "hard streaks" that segment the reservoir.

### **Fracturing**

Natural fractures identified in core samples and borehole image logs contribute little to porosity but serve as important pathways for fluid flow, making them key drivers of permeability. At the core scale, the Tuba Formation does not exhibit extensive fracturing—no widespread open fracture networks were observed. However, small micro-fractures and hairline cracks are present, particularly within tight intervals. Petrographic thin sections occasionally reveal these micro-fractures cutting across grains and cement; some are sealed with calcite or dolomite, while others remain open.

### **Diagenetic Variability**

The range of diagenetic alterations in the Tuba Formation is characterized by the type of diagenetic processes varying according to the primary depositional framework. Coarser-grained facies such as skeletal grain and packstones experienced extensive textural modification due to multiple diagenetic events under varying thermal and burial conditions. These processes include early-stage mechanical compaction, fabric-selective dissolution, various episodes of cementation, and late-stage fracturing. In particular, rudist grainstones, characterized by well-connected framework grains, show minimal pore-filling cementation. Instead, they were more affected by compactional fractures and localized mud seams. The preservation of depositional porosity and enhanced pore connectivity due to selective dissolution contributed to relatively higher permeability in these facies.

In contrast, medium-grained facies like argillaceous skeletal packstones and wackestone exhibit more intense diagenetic overprinting. The compaction of their mud-rich matrix resulted in multiple sets of clay seams. Concurrently, carbonate dissolution during compaction led to centimeter-scale nodular cementation, clearly visible in core samples. Continued burial compaction produced various generations of fine cracks within these nodular cements. Although pore-filling calcite cement partially occluded these cracks and residual primary pores, some retained limited connectivity. Neomorphic recrystallization of micrite to micro spar calcite further reduced pore throat sizes, lowering permeability. Dissolution generated moldic and vuggy macro pores in certain intervals, and these pores were mainly isolated and poorly connected.

Finer facies, such as argillaceous mudstones, preserved matrix-supported depositional textures and were dominated by neomorphic calcite replacement. These intervals are also characterized by discontinuous compactional clay seams, which, along with associated cementation, led to almost complete occlusion of pore space. As a result, these facies exhibit very low porosity and permeability, functioning as tight zones and potential flow barriers within the reservoir framework. (Sunil, 2024)

### **Diagenetic Zonation**

Based on core observations, thin-section petrography, and petrophysical data, three distinct diagenetic zones are identified within the Tuba Formation.

### Moderate Diagenetic Zone (MDZ)

This zone predominantly occurs in grain-supported facies, especially skeletal grainstones and clean skeletal packstones, roughly corresponding to the lower stratigraphic interval of the Tuba Formation. The MDZ is marked by well-preserved interconnected primary porosity and extensive dissolution (Fig. 2b), with minimal calcite cementation. Porosity typically exceeds 15%, and permeability ranges from tens to over 20 millidarcies (table 1). Intergranular and moldic types dominate pore systems. This zone hosts the most favorable reservoir flow units, although dolomitization is heterogeneous within this zone, overall pore connectivity remains high, making the MDZ the most productive interval in terms of reservoir performance.

**Table 1—Summary of diagenetic zones and their associated reservoir characteristics in Well-B, Tuba Formation.**  
The table highlights variations in porosity, permeability, facies types, and dominant diagenetic features across Low, Moderate, and High Diagenetic Zones, and corresponding Flow Units (Low-, Moderate-, and High-Performance Units).

| Diagenetic Zone | Flow Unit | Depth Interval (ft.) | Thickness (ft.) | Avg. Porosity (%) | Avg. Permeability (mD) | Principal Facies                 | Dominant Diagenetic Features                         |
|-----------------|-----------|----------------------|-----------------|-------------------|------------------------|----------------------------------|------------------------------------------------------|
| LDZ             | LPU       | 7020–7022            | 2               | 20.35             | 0.84                   | Argillaceous Skeletal Wackestone | Isolated macro pores and dissolution vugs            |
| MDZ             | HPU       | 7022–7024            | 2               | 26.27             | 21.67                  | Skeletal Grainstone/ Packstone   | Abundant dissolution-enhanced secondary porosity     |
| HDZ             | MPU       | 7024–7029            | 5               | 18.70             | 3.07                   | Skeletal Wackestone              | Nodular calcite cement, neomorphism, moldic porosity |
| LDZ             | LPU       | 7029–7042            | 13              | 3.91              | 0.42                   | Argillaceous Skeletal Wackestone | Neomorphic and pore-filling cements                  |
| HDZ             | MPU       | 7042–7045            | 3               | 19.24             | 2.19                   | Skeletal Wackestone              | Excessive pore filling cement, Moldic pores          |

### High Diagenetic Zone (HDZ)

This zone includes the transition between grain-rich and mud-rich facies, dominantly located in the upper part and less towards the lower part of the Tuba Formation. It is characterized by intense heterogeneous cementation, compaction, and pressure-solution, with limited dissolution and partial dolomitization. Porosity ranges from 8% to 15%, and permeability typically falls in the 1–10 mD range (table 1). Facies include intraclastic wackestone, argillaceous packstones, argillaceous skeletal wackestone, and skeletal packstones. Nodular calcite cementation is dominant, often cracked due to compaction. While moldic, fracture porosity is present, it is frequently isolated due to fabric-selective dissolution and cement occlusion. The HDZ is highly heterogeneous, consisting of alternating permeable and impermeable streaks, often with thin (1–3 ft.) cemented layers acting as intra-zone baffles. As a result, this zone cannot be treated as a single continuous flow unit but rather as an interbedded package of moderate-quality reservoir and minor barriers.

### Low Diagenetic Zone (LDZ)

This zone is dominated by fine-grained facies such as argillaceous mudstone and shale. These facies exhibit poor reservoir properties, with less than 10% porosity and permeability below 1mD (table 1). The LDZ represents a regional flow barrier or internal seal, seen mainly in the middle part and less in the upper part of Tuba Formation.

## Petrophysical Characteristics and MICP Trends

The impact of the aforementioned diagenetic variability on rock properties is reflected in core porosity and permeability data and in MICP-derived parameters. Core analysis results show a wide scatter when all the data from the Tuba Formation are plotted, but clustering becomes evident when colored by diagenetic facies (fig 3):

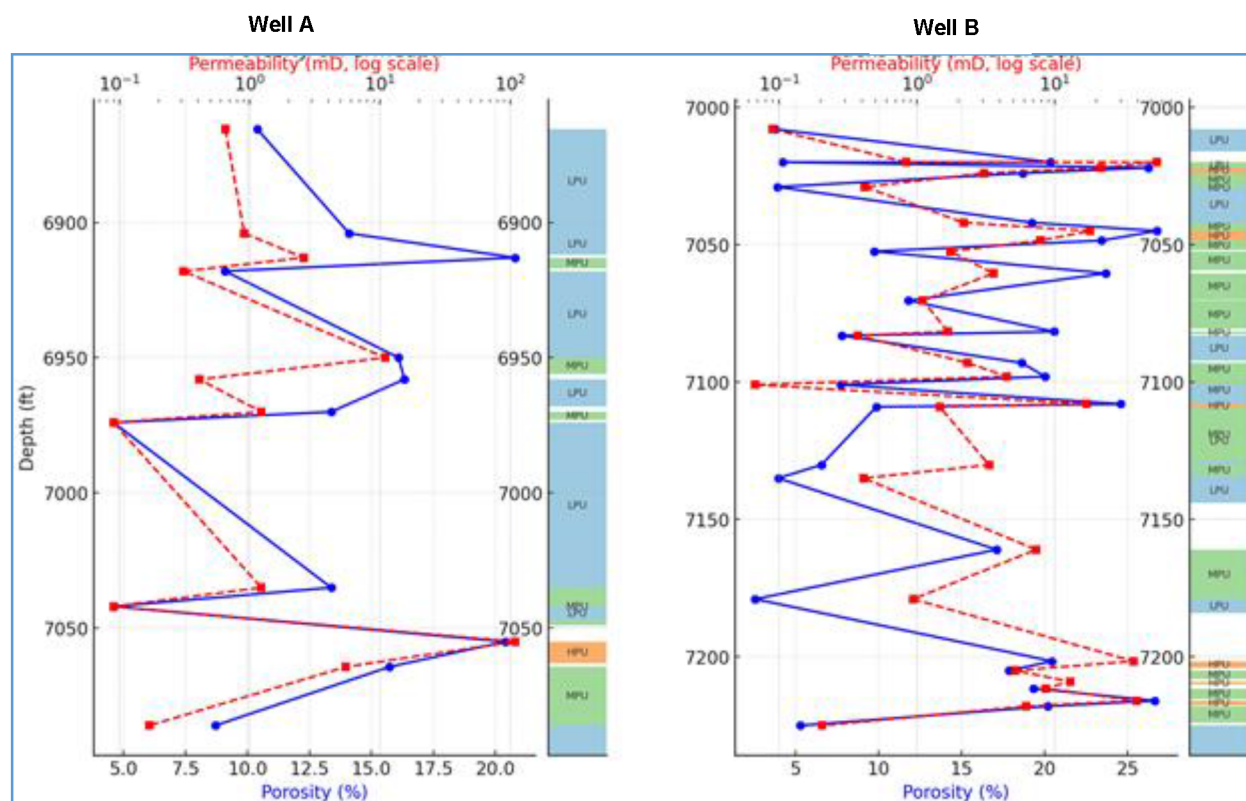
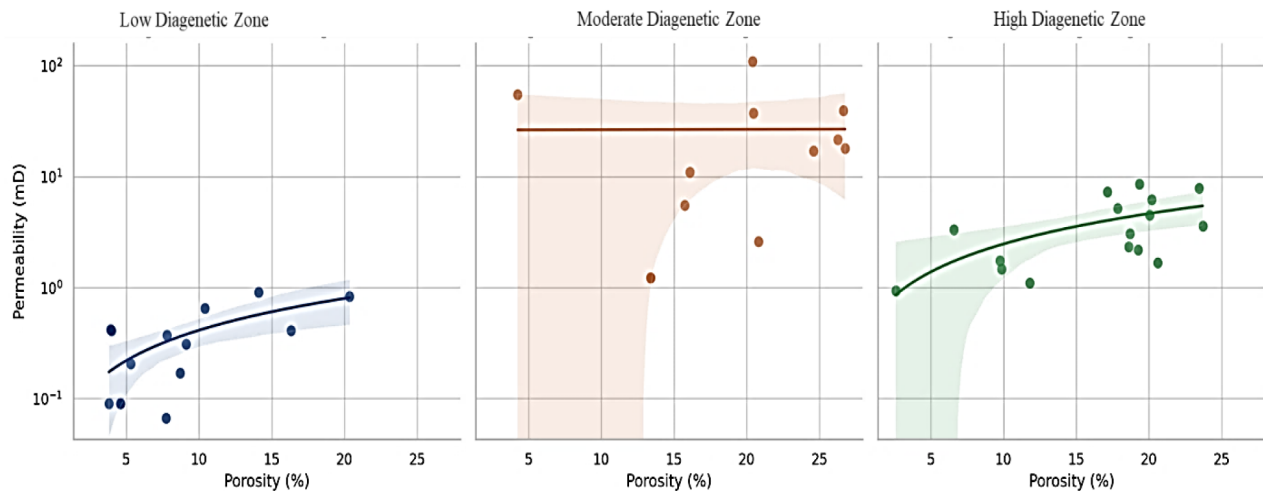


Figure 3—Zonation-based variation in porosity and permeability across the Tuba Formation

The porosity–permeability profiles highlight strong heterogeneity linked to diagenetic zonation. Samples from dissolution-zones, typically MDZ, occupy the high end of the spectrum, with helium porosities of 20–30% and air permeability's commonly between 1–30 mD. In a cross-plot of porosity vs. permeability, these samples define a trend of high perm at moderate porosity, reflecting efficient pore networks (large, well-connected throats). Samples from cemented or compacted zones (LDZ or cemented layers in MDZ) show low porosity and ultra-low permeability. Many have porosities in the single digits and permeability's below 0.1 mD. Some cluster around 10–15% porosity and 1–10 mD permeability. This reflects the microporous nature of any remaining pore space and the lack of connected pathways. Even samples with slightly higher porosity (~10%) in this category can have permeability's still in the 0.1–1 mD range, indicating that pore throats are extremely constricted (likely sub-micron). Such samples align with known "barrier" facies and confirm their negligible flow contribution.





**Figure 4—Zone-wise cross plots of porosity (%) versus permeability (mD), with individual linear regression trend lines for each Diagenetic Zone (Low, Moderate, and High).**

Samples from intermediate, partially altered zones (HDZ baffle units) are more variable. Some cluster around 10–15% porosity and 1–10 mD permeability. Others with isolated moldic porosity show higher porosity (~15–18%) but still low permeability (~1 mD or less), representing unconnected porosity. They occupy a middle band between the high-perm dolomite trend and the tight cemented trend when plotted. The scatter here is greatest because these rocks have a mix of pore types. A key observation is that permeability in these samples is often an order of magnitude lower than one would predict from porosity alone if assuming a well-connected pore network. This underperformance is directly tied to diagenetic factors

### 3D Model Insights: Reservoir Layering and Compartmentalization

Using the well data and interpretations, a conceptual 3D static model of the Tuba Formation was built to visualize the spatial distribution of facies and diagenetic zones. Although limited by the number of wells and the inherent uncertainties in lateral facies continuity, this model provides a useful depiction of how diagenetic heterogeneity might be arranged in three dimensions and how it could compartmentalize the reservoir.

#### Layering and Lateral Continuity

The Lower Diagenetic Zone (LDZ) is mapped as a laterally continuous tight layer across the field area – effectively a cap rock within the reservoir. This is consistent with a regionally deposited mudstone/shale-rich interval, forming a blanket with minimal reservoir quality. All wells penetrate this unit and show similarly low porosity readings on logs, suggesting its continuity. The High Diagenetic Zone (HDZ) exhibits a more complex internal architecture. The model indicates that within the middle depths of Tuba, multiple thin layers of better reservoir quality grade laterally into tighter facies. The cemented wackestone layers are intercalated with these, which can often be correlated as semi-continuous baffles. The model's Moderate Diagenetic Zone (MDZ) is represented by laterally extensive grainstone packages in the lower Tuba.

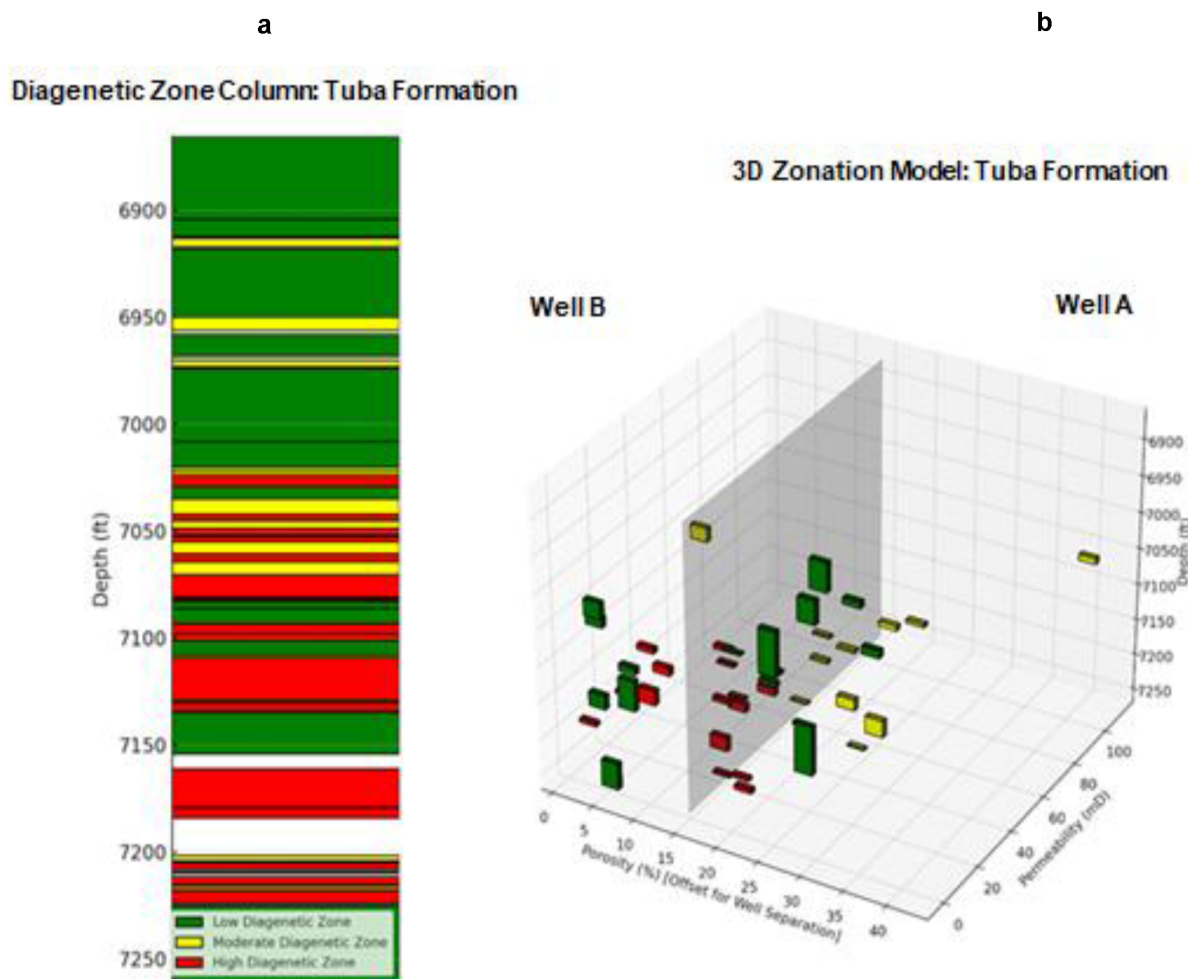


Figure 5—Stratigraphic column and zonation model representing Tuba Formation. Stratigraphic column representing the Tuba Formation compiled from well B and A, illustrating vertical distribution of diagenetic zones with depth. Each colored interval denotes a diagenetically distinct zone: Green- Low Diagenetic Zone, Yellow – Moderate Diagenetic Zone, Red - High Diagenetic Zone, White – Gaps in cored wells. Zonation model integrating well A and B, illustrating diagenetic heterogeneity in porosity–permeability–depth space. Each colored cuboid represents a distinct diagenetic zone: green for Low, yellow for Moderate, and red for High Diagenetic Zones.

## Discussion

### Influence of Diagenetic Processes on Reservoir Quality and Heterogeneity

The results of this study demonstrate that diagenetic processes have been critical drivers of reservoir quality variation in the Tuba Formation. While depositional facies created the initial architecture of potential reservoir and non-reservoir layers, diagenesis has both exacerbated and modified that architecture, in some cases creating high-permeability conduits in rocks that originally had modest porosity, and in other instances destroying porosity in rocks that originally were good reservoirs. These findings reveal that heterogeneity in the Tuba Formation exists at multiple scales: at the macro scale, it is controlled by depositional cyclicity, while at the micro scale, it is influenced by localized diagenetic overprints

### Reservoir Management and EOR Implications

The heterogeneity induced by diagenesis has direct consequences for Enhanced Oil Recovery strategies. In a homogeneous reservoir, a waterflood or gas flood can sweep oil uniformly; in Tuba's case, however, the presence of high-permeability streaks adjacent to low-permeability (table 2) layers means that any injected fluid will preferentially flow through the path of least resistance – i.e., the dolomitized, vuggy channels – and

bypass the tight oil-bearing zones. This phenomenon of channeling or fingering leads to early breakthrough of injectant in producers and poor overall sweep efficiency. The heterogeneous multi-thickness layers of tight carbonates require a multi-pronged approach:

**Table 2—Recommended Enhanced Oil Recovery (EOR) strategies for the Tuba Formation, organized by diagenetic zonation. Each strategy targets zone-specific petrophysical attributes—such as porosity, permeability, and cementation—and addresses sweep inefficiencies caused by reservoir heterogeneity. The practices are based on published case studies and field-proven applications in heterogeneous carbonate systems.**

| Diagenetic Zone              | Characteristics                        | Key EOR Strategy                                         | Targeted Benefit                                    | References                                                                 |
|------------------------------|----------------------------------------|----------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------|
| <b>LDZ (Low)</b>             | Tight, calcite-cemented, low flow      | Pinpoint acid fracturing / Hydraulic fracturing(tested ) | Create conductivity in tight zones Dissolve cements | <a href="#">Ranjan,2018</a> ; <a href="#">Yang,2023</a>                    |
| <b>MDZ (Moderate)</b>        | Heterogeneous, dolomitized grainstones | Gel conformance control Selective injection              | Divert injectant to HDZ Delay breakthrough          | <a href="#">Seright, 2012</a> ; <a href="#">Zaitoun &amp; Tabary, 2012</a> |
| <b>HDZ (High)</b>            | Moldic porosity, moderate permeability | Gel diversion /Acid stimulation                          | Access isolated baffles, Improve connectivity       | <a href="#">Ranjan, 2018</a>                                               |
| <b>High porosity Streaks</b> | Vuggy, high-perm conduits              | CO <sub>2</sub> or WAG injection                         | Control channeling Improve vertical sweep           | <a href="#">Dai, 2013</a>                                                  |
| <b>Compartmental Zones</b>   | Vertically isolated flow units         | Zonal flooding Chemical EOR Massive fracturing           | Tailor strategy per unit Maximize net recovery      | <a href="#">Zheng, 2010</a>                                                |

### Zone-specific Injection Strategies

To enhance injection efficiency and sweep performance in the Tuba Formation, a targeted approach is recommended over conventional bulk flooding. Selective perforation or completion of injectors in both high- and low-permeability zones, combined with controlled injection rates, allows for better management of reservoir heterogeneity. In moderate diagenetic zones (MDZ)—typically dolomitized or partially cemented grainstones—polymer gels or gel micro particles can be applied to restrict flow through dominant pathways. This conformance control promotes better injectant redistribution into high diagenetic zones (HDZ), which often possess moderate permeability and complex, underswept pore structures. Meanwhile, the low diagenetic zones (LDZ), being tight and heavily cemented, are less suited for gel treatment but may respond effectively to targeted acid stimulation or hydraulic fracturing. ([Fig 6](#))

In the heterogeneous Tuba Formation, invoking polymer gel or microgel for conformance control can selectively reduce permeability in high-flow MDZ grainstone layers, thereby forcing injectant into HDZ baffles that are usually bypassed. This aligns with field-proven applications in fractured carbonate systems ([Seright, 2012](#); [Zaitoun & Tabary, 2012](#)), where gel treatments have led to 10–15% incremental recovery within months. Our diagenetic zonation model identifies MDZ layers as ideal for gel placement, with flow diversion to HDZ and selective stimulation of tight LDZ mudstones where necessary.

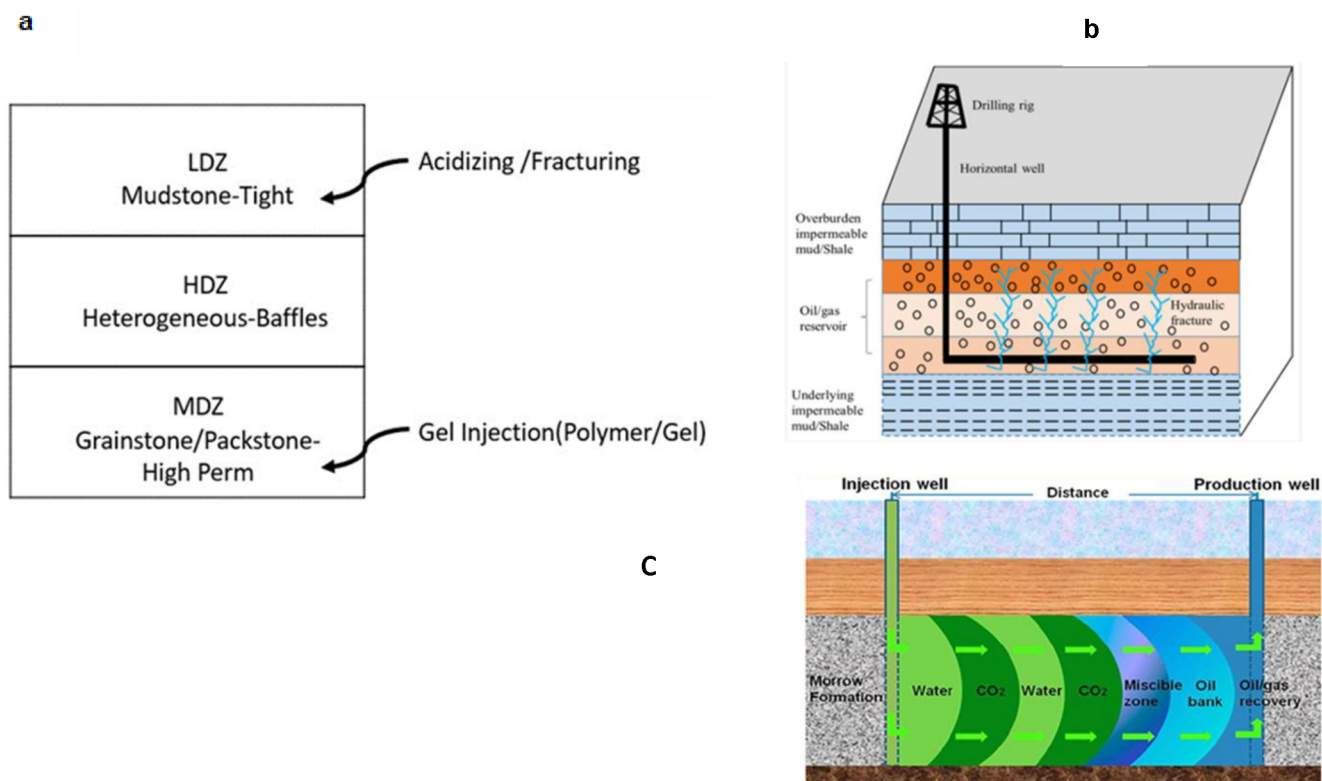


Figure 6—EOR planning guided by established conceptual models in literature. This schematic is conceptually adapted from field implementations of polymer/micro gel conformance control in heterogeneous carbonate reservoirs (Zaitoun, 2012; Seright, 2012). b. Conceptual schematic adapted from hydraulic fracturing workflows in Middle Eastern tight carbonate formations (Yang, 2023; Ashena, 2022), showing fracture zones planned through LDZ and HDZ intervals targeting connectivity across tight and heterogeneous layers. c. Conceptual depiction of CO<sub>2</sub>-WAG injection in a horizontally layered reservoir. Alternating water and CO<sub>2</sub> slugs target high-permeability zones (e.g., MDZ grainstones), reduce early breakthrough, and drive CO<sub>2</sub> into tighter units (e.g., HDZ baffles), enhancing vertical sweep—adapted from Dai. (2013).

### Hydraulic Fracturing of Tight Intervals

Hydraulic fracturing (propped fracturing) of these intervals can create artificial conduits that connect the tight matrix to the wellbore and potentially penetrate through barrier layers to connect isolated zones. The results indicate that the most suitable candidates for fracturing would be the cemented mudstone layers and compact wackestone in the MDZ and HDZ.

### Matrix Acidizing / Acid Fracturing

Carbonate reservoirs uniquely allow the possibility of acid stimulation to dissolve rock and enlarge pores. In Tuba's context, acidizing the tight intervals in the HDZ can reopen or enlarge pore throats. The efficacy will depend on acid contact, where techniques like pinpoint acidizing come in. It was demonstrated in a recent field trial (table 2) that pinpoint coiled-tubing conveyed acid fracturing, where each stage is tailored to a specific tight zone, led to significant productivity improvement in a horizontal well in the Tuba carbonate. This method treated each diagenetic "barrier" streak individually with an acid fracture, significantly increasing the number of effectively stimulated zones compared to conventional approaches (Ranjan, 2018). The success of that approach validates this study's implication: targeted acid-fracture stimulation is a key to unlocking the oil in diagenetically tight zones. By focusing acid where calcite cement is abundant, it will preferentially dissolve the cement and create new pore space or channels (table 2).

With pinpoint technology, stimulation is delivered precisely to chosen intervals using a drill pipe and pinpoint tool assembly. This avoids treating non-productive or low-quality rock, ensuring acid is only placed where it can enhance permeability and productivity.



### Enhanced Gas Injection (CO<sub>2</sub> or Miscible) in High-Permeability Streaks

For the high-permeability zones, consider miscible gas injection like CO<sub>2</sub> flooding, which can efficiently displace oil due to multi-contact miscibility and has the side benefit of lowering interfacial tension. CO<sub>2</sub> could potentially penetrate slightly into tighter pores than water due to lower viscosity and some rock-dissolving capability (carbonic acid formation). However, CO<sub>2</sub> would still race through the most prominent pores without conformance control and bypass others. A water-alternating-gas (WAG) approach (Fig. 6c) could help mitigate this by periodically blocking CO<sub>2</sub> flow with water. The heterogeneity information from the study model could be used in a simulation to design an optimal WAG ratio and slug size to improve vertical sweep.

### Segregated Recovery Mechanisms

Given the compartmentalization, another strategy is to produce different compartments with different methods. For example, the bottom grain-dominant zone might respond well to waterflood, the middle zone might require chemical flooding (e.g., a surfactant or polymer flood to push oil out of small pores) (Zheng, X, 2010). The fact that HDZ is extremely cemented suggests it might behave like a separate reservoir – possibly it could even be developed via horizontal wells that are massively fractured. Meanwhile, the underlying better zones can be water flooded concurrently.

### Implications for EOR and Reservoir Management

The proposed workflow has direct application in EOR by integrating diagenetic zonation with reservoir quality and flow characterization. It enables selective stimulation of high-permeability intervals, improved conformance control in thief zones, and zone-specific recovery strategies tailored to facies and diagenetic attributes. This structured approach provides a practical framework for reducing sweep inefficiencies and guiding field-scale EOR implementation. By accurately identifying diagenetically controlled flow units and their petrophysical behavior, the workflow allows targeted EOR strategies that directly address reservoir heterogeneity—one of the major challenges in carbonate systems. For example, high-permeability streaks can be managed through conformance control measures such as polymer or gel placement, while zones with dissolution-enhanced porosity but limited connectivity can be selectively stimulated to improve fluid flow. The predictive 3D zonation model further supports optimized injection planning by distinguishing thief zones from effective storage zones, ensuring more efficient displacement. Overall, the workflow not only advances geological understanding but also translates into practical, field-relevant improvements in reservoir management and recovery performance.

## Conclusion

This study highlights the critical influence of diagenetic processes on reservoir quality and heterogeneity in the Tuba Formation of Kuwait. While depositional facies laid the foundation for reservoir architecture, diagenesis significantly modified flow behavior, enhancing or destroying porosity through cementation, dissolution, dolomitization, and compaction. High-quality intervals are associated with preserved or secondary porosity, while cemented mudstones with ultra-low permeability dominate tight intervals. Diagenetic zonation divides the reservoir into high-flow, baffle, and barrier units, creating vertical and lateral compartmentalization, as captured in the 3D conceptual model. These findings carry important implications for Enhanced Oil Recovery (EOR) strategies—highlighting the need for targeted stimulation, effective conformance control, and zone-specific injection approaches to minimize bypassed oil and enhance recovery efficiency.

Eventually, this study offers a comprehensive framework that integrates geological and petrophysical insights into field development planning, delivering practical guidance for optimizing production in the Tuba Formation and other carbonate reservoirs marked by complex diagenetic overprints, offering practical

solutions for maximizing productivity in the Tuba Formation and other diagenetically complex carbonate reservoirs.

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